Title: Quantum Computing and the Future of Cryptography

Abstract:
Quantum computing poses a significant threat to modern cryptography. This paper explores the fundamen

1. Introduction
Modern secure communication relies on the difficulty of factoring large prime numbers, a problem RSA enc

2. Quantum Mechanics in Computing
Quantum computers use qubits, which can exist in a state of superposition (representing 0 and 1 simultane

3. Shor's Algorithm
Developed by Peter Shor in 1994, this quantum algorithm can find the prime factors of an integer exponen

4. Post-Quantum Cryptography
To prepare for Q-Day, NIST has been evaluating post-quantum cryptography (PQC) algorithms. Lattice-ba

Conclusion:
The transition to quantum-safe encryption is not optional. Organizations must begin adopting lattice-based